

A Data Scientist who views mathematics as the formal language of imagination. I specialize in bridging the gap between creative intuition and algorithmic rigor, transforming abstract ideas into structured, high-impact models. By conceptualizing the underlying mechanics of machine learning before the first line of code is written, I build solutions that are as logically sound as they are innovative.

SKILLS

Programming languages	Skilled in Python and SQL , with a working knowledge of Java and Scala .
Tech stack	dbt, Vertex AI, BigQuery, PostgreSQL, Ray, Elasticsearch, Grafana, Spark, Airflow
Statistical	Real-world experience in Causal Inference, Time-series model, Tree-based model. Understand academic concepts of Graph, Learning to Rank, LLM.

EXPERIENCE

Data Nest **11/2025 – Present**
Data Scientist

- **Lead generation and lookalike model:** Analyzed complex business processes to define strategic Proxy Labels. Design evaluation metric of lookalike model to ensure lead quality. Build end-to-end training pipeline of lookalike model.
- **Location Feature Engineering:** Designed and built an end-to-end spatio-temporal feature system converting raw telecom logs (Voice, SMS, HLR/VLR) from VNPT & Viettel into subscriber behavioral features for credit scoring. Replaced static cell mapping with a 6-month dynamic join window, resolved Vietnam's administrative boundary restructuring, and deployed automated null-rate & cell-count drift monitoring. Feature architecture: 3-layer granularity (Cell/District/Province), 4 behavioral profiles (Home/Work/Weekend/All), spatial metrics including Radius of Gyration, Spatial Entropy, Earth Mover's Distance, and Meta RWI enrichment via spatial join. Improved Gini across 5 of 6 credit scoring models on VNPT data, peak gain **+3.12% Gini**.
- **HDBank BNPL Model Enhancement:** Identified and fixed a data leakage in the legacy evaluation framework by enforcing temporal train/test split. Discovered the Inter-Arrival-Time (IAT) feature during Service Number Detection R&D — a behavioral signal capturing call-interval patterns. Used **PySpark ML** to evaluate a 200+ feature inventory and select the top 50 by importance. Integrated HDBank credit card data into BNPL training set to capture bank-specific default behavior. Final model achieved **+1.76 Gini** uplift on BNPL portfolio and **+2.17 Gini** on CC portfolio, narrowing the gap with the legacy CC-specialist model to under 2 points.

Deliverree on-demand logistics **10/2024 – 11/2025**
Data Scientist

- **Dynamic Pricing (built from scratch):** Achieved highest increment of GTV recorded is 40% in A/B testing
 - **Price-demand model (causal):** From zero to production — set the goal, collected/cleaned event data, built a model to measure how price changes affect demand, ran experiments, and deployed with **Ray Job**. Added shadow rollout and monitoring.
 - **Potential demand model:** Changed the forecast target from 1 hour to 3 hours (with upsampling) after checking the uncertainty in estimates. This made pricing more statistically reasonable.
 - **Supply model:** Designed a new supply model to replace the old “not-fulfilled” model. Prove by causal inference that the effect of price is not significant on new target in case of our business. Relax the assumption about causality of supply.
 - **Price safety:** Proposed and tested a statistical guardrail to catch market reactions and adjust prices if needed, so prices don't swing too far in case the pricing system go wrong.
 - **Engineering:** Built the pricing API with **Ray Serve**, trained models on **Google Vertex AI Pipelines**, and set up **Grafana** dashboards for monitoring, design model output pipeline on Postgresql to ensure the availability of data.
- **Driver matching system (enhancement):**
 - Define and collect data from the driver portfolio.
 - Build error analysis tree: point out the cohort that have low accuracy
 - Optimize training flow: re-write sampling method for lower the training time from 12 hour to under 2 hour.

Thesis : Speech Enhancement using deep-learning on edge device **01/2023 – 06/2023**

- Applied DCCRN model with feature fusion and similarity-preserved knowledge distillation to compress a speech enhancement model for edge deployment. Implemented complex-number neural networks in PyTorch and optimized multi-GPU training with PyTorch Lightning. **Grade: 9.1/10 [thesis]**

+84 969-455-608
Tan Phu, Ho Chi Minh city
congduyvu.2103@gmail.com

Vu Cong Duy

Data Scientist/ AI Engineer

github.com/tonyvu1289
linkedin.com/in/vucongduy
kaggle.com/congduyvu

Timo Digital Bank

04/2023 – 06/2024

Data Scientist

- Developed an expense prediction model by redefining the target variable and implementing a data pipeline with **dbt** and **Airflow**. Delivered insights through dashboards on **Holistics** and personalized customer engagement using **Mandril**.
- Enhanced transaction purpose classification by applying **rule mining**, **Elasticsearch** (TF-IDF, Bayesian clustering), and advanced evaluation with **EvidentlyAI**. Deployed the improved model as an API service using **SpringData**.
- Designed and implemented graph algorithms (**DFS**, **BFS**, **heuristic growth**) on 4TB of user interaction data with **PySpark**, uncovering customer behavior patterns that informed UI/UX improvements in money transfer and adding contact flows.
- Referral : Ngoc Anh - Head of Data at Timo

Vietnam International Bank(VIB) Customer Data Platform

09/2022 – 01/2023

Vendor Data Scientist

Bluebolt software

- Scoped DS use cases (churn prediction, spending regression, recommendation) by auditing data across VIB's card, digital, and marketing departments. Assessed data quality and built dashboards on SAP to track DS metrics for the BA team.

EDUCATION

Bachelor of Computer Science, Specialized in Data Science

HCM University of Science - VNU, GPA: **8.46/10**

04/2019 — 08/2023

Master of Data-science HCM University of Science - VNU

08/2024 — Present